

Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion

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Editor Comments

- Please note, references suggested by reviewers should only be added if it is relevant to the article and makes it more complete. There is no mandate to include these specific references in your revision
- Please note, Data Descriptors should not include significant results, conclusions, discussion or data analyses. The goal is to share data for others to use, with the details in the paper intended to provide transparency (how data were collected or validated) or practical help (how to find the data, what variables/column headings mean, etc.)
- Please remove the section numbering.
- Please check the Background and Summary section to ensure it is in scope and does not contain any general results, analyses, conclusions or facts relating to what the data shows. The section should be limited to brief descriptions of the motivations of the project and any relevant prior art. Data Descriptors do not analyse data or present findings, as the goal is for others to analyse the data for themselves.
- Please add a data citation for the dataset to the reference list using these instructions (note that a DOI URL should be used, not any other link). Please add the reference number to wherever the dataset is mentioned in the text - the main position should be the first part of the Data Record in a sentence describing where the data has been deposited.

Response to Editor

Thank the Editor for their thorough feedback and address each comment individually below.

The references suggested by the reviewers have been reviewed, and the relevant ones have been added to the manuscript.

We understand the constraints regarding the disclosure of results, conclusions or data analyses. For this reason, we have not been able to incorporate the reviewers' requests on this point.

Section numbers have been removed from the manuscript; references to sections now use their names instead.

The Background and Summary section has been revised to remove any general results, analyses, conclusions, or facts.

The citation of the figshare repository has been updated to follow the format provided.

Reviewer 1

The paper presents a dataset of a single-segment tendon-driven continuum robot under several sensing modalities. The inclusion of pose and contact force information on an external object is interesting.

Comment 1.1 *I consider that the paper needs a general review of how to present the developments achieved. Consider including diagrams, pictures, or flowcharts to help readers understand the process described. For example, Figure 1 is not clear what the author refers to by the contact wand; g-fix is not clear what it means or how it is used. Maybe can include a Figure in the text: "The four actuator-sensor pairs are labeled 1-4, corresponding to tendons aligned with the x , y , $-x$, and $-y$ directions"*

Response to Comment 1.1 We thank the reviewer for raising this point on clearness. Many sections were changed to address these readability concerns. Concerning g_{fix} this homogeneous transformation represents the fixed rotational and translational offset between the mocap frame attached to wand and the HEX 12 sensor frame. In detail, applying g_{fix} to the wand frame allows to obtain the position and orientation of the HEX 12 (in which the measured wrench is expressed). We understand the confusion about the labeling of the actuator and sensor pairs. The sentence alone is not sufficient to clarify association between the label and the tendon they apply to.

Changes made to the article The Methods and Usage Notes Sections have been updated to clarify the acquisition and postprocessing pipelines. Figure 2 was modified to clarify the ordering of the actuator-sensor pairs on the robot base frame and the frames associated with the contact wand and HEX 12 sensor. The description of g_{fix} has been updated for clarification.

Comment 1.2 *The author mentions "Together, these design choices ensure that each recorded quantity has a clear physical correspondent in the governing equations of tendon-driven continuum robots". However, in the text it is not clear that the dataset fulfills any equations of tendon-driven continuum robots.*

This response may cause confusion that the design is mainly for the Cosserat rod model. I also add a similar comment in the paper.

Response to Comment 1.2 We agree the original phrasing overstated this point. We have reworded the sentence to specify exactly which quantities are meant and their role in the modeling frameworks used elsewhere in the paper: the tendon tensions serve as the actuation input, the base wrench provides the boundary condition, and the disk poses and reconstructed backbone shape describe the robot's configuration.

Changes made to the article Sentence revised as described above, in Section Continuum Robot Platform.

Comment 1.3 *Using the sensor information as Opti-Track, kinematic information can be obtained; however, it is not clear how a forward dynamics simulation based on the Geometric Variable Strain (GVS) approach is used. What kind of information is obtained ?. It is convenient to compare the performance of the dynamic model with that of a single-segment tendon-driven continuum robot.*

Response to Comment 1.3 We understand the reviewer concern. The forward dynamics simulation is used to correlate measured tendon tensions with the measured torque at the base, since a direct comparison of these quantities is not possible without accounting for the physical behavior of the system. To this aim, tendon tensions are fed to a forward dynamics simulation based on the GVS which computes the wrench at the base of the robot. We have extended the cross-validation sentence in Background & Summary to name this check.

Changes made to the article Extended the cross-validation sentence in Background & Summary to name the dynamics-model-based consistency check reported in Technical Validation.

Reviewer 2

The manuscript entitled "Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion" represents a very important contribution to the field. There is no doubt that such a dataset is missing

and would accelerate the development of improved control models. Nevertheless, I have two concerns that might be addressed by the authors. Both of my concerns revolve around the topic of embodied intelligence. While I agree that a dataset is missing to research and develop better control models, I also see the challenge of an intertwined hardware/software issue. Continuum robots, like many biological systems, are often designed for specific tasks, where the task guides the design.

Comment 2.1 *First concern: In the introduction, the authors potentially over-state the need for a dataset in continuum robotics. Or at least, they neglect, the above mentioned, very important, aspect that make controlling such robots challenging: embodied intelligence or morphological computation. Unlike in the other fields, mentioned in the intro, the motion in the present case is inherently coupled to the construction material and hardware design of the robot. I suggest to rethink the paragraph maybe even talking about the contribution of hardware and environment to control.*

I don't think this dataset can be used for comparing control algorithms. In addition to agreeing on the generalization limitation due to material and hardware, I think we should also make it clear that the main target applications of this dataset is modeling, system identification and state estimation.

Response to Comment 2.1 The original framing risked overstating the analogy with datasets such as KITTI, Dex-Net, and EuRoC, whose target tasks are largely decoupled from the specific hardware that generated them. Indeed, for tendon-driven continuum robots, dynamics is inherently coupled to the mechanical design and material properties of the platform, an instance of embodied intelligence.

Changes made to the article We added a paragraph to the Introduction acknowledging this hardware coupling, and extended the "Limitations of applicability" paragraph in the Usage Notes with a companion sentence tying the dataset's generalization scope to the same point.

Comment 2.2 *Second concern: The present design was deliberately selected that the tendons do not contact the motor case for better measurement quantities. I agree, but I am wondering whether the whole robot design requires additional force sensors at the contact points of tendons and the inner surfaces of the equally spaced disks. By shortening one cable, the last disk is pulled towards the first disk and the cable makes contact to the outer part of the inner surface, where the cable is routed through. I suggest to comment on this fact and either explaining, why this measurement quantity is not needed or adding a statement for the limitations of the current design or dataset.*

Since the reviewer explicitly suggest to add a statement, we may just add one brief sentence in the paper, saying that measuring the tendon tension alone is a common practice in the community?

Response to Comment 2.2 Although measuring the tendon-disk contact force is interesting in principle, it carries two distinct problems. First, adding a force sensor at each disk tendon guide would significantly complicate the design and make the robot's mechanical properties more difficult to identify, as the cables routed along the backbone would alter stiffness, inertia, and damping in a non-homogeneous, arc-length-varying way. Moreover, tendon-disk contact is deliberately minimized to comply with classical modeling assumptions and to reduce friction; introducing a force sensor at that location would increase the contact surface or introduce substantial design challenges. Second, because this quantity is technically difficult to measure, mainstream modeling approaches rely on cable tension alone. Thus, even if this quantity had been provided, it would have had little to no practical use.

Changes made to the article No change was made to the article; the design rationale explaining why tendon-disk contact force is not measured is provided in the response above.

Comment 2.3 *Additionally, I found one typo, the authors might want to correct. Intro: please correct typo "between heterogeneous force measurements" instead of "between heterogeneous force measuraments"*

Response to Comment 2.3 We thank the reviewer for the careful reading. The typo has been corrected.

Reviewer 3

General Comments *This paper presents a complete, well-validated multi-modal dataset for single-segment tendon-driven continuum robots, filling a clear gap in open benchmark resources for continuum robotics. The authors integrate six synchronized sensing modalities and split the dataset into quasi-static, dynamic trajectory, and contact interaction subsets, covering static identification, dynamic modeling, learning-based estimation and contact analysis. The experimental platform is fully documented with*

commercially available hardware, and both raw and uniformly processed data are shared on Figshare with open acquisition/post-processing code. Rigorous cross-modal validation including kinematic matching, force consistency checks, sensor noise characterization and temporal synchronization analysis thoroughly verifies dataset reliability for continuum robot. The manuscript is well-structured and technically sound, and only minor revisions are required before acceptance. Here are some suggestions:

Comment 3.1 *Add dedicated discussion about the generalization limitation of continuum robot datasets and corresponding improvement strategies to enhance the work’s broader impact: It is widely acknowledged that continuum soft robots face an inherent trade-off between abundant compliant degrees of freedom and precise controllability, making data-driven modeling an indispensable solution. Compared with other soft robot variants, tendon-driven continuum robots benefit from relatively analytical kinematic frameworks, which partially mitigate this contradiction. Nevertheless, intrinsic material uncertainties such as viscoelastic creep, hysteresis and manufacturing deviations persistently degrade model transferability. Even for two structurally identical continuum robot prototypes trained on the same dataset, data-driven models often fail to generalize directly—new data collection and retraining are mandatory for each individual platform, severely limiting the reproducibility and reusability of open datasets and codes across laboratories. The authors are required to supplement targeted discussion of this critical challenge in the manuscript, and elaborate concrete approaches to improve the universality and cross-platform generalization of the presented dataset, platform and released codes for future research communities.*

Response to Comment 3.1 We thank the reviewer for raising this point. Even a structurally identical replica of the robot used here is expected to differ in backbone viscoelastic creep and hysteresis, marker/sensor placement tolerances, and tendon pretension, all of which are unit-specific and cannot be characterized once and for all by a single dataset. We added a dedicated discussion of this limitation to the Usage Notes, clarifying what this dataset already provides toward mitigating it: the quasi-static subset already serves to characterize platform-specific physical properties (e.g. the backbone’s equivalent bending rigidity, used elsewhere in this manuscript), and the same protocol applied to a new unit gives good grounds for identifying its own residual, unmodeled physics. We are explicit that fully solving cross-platform generalization is beyond the scope of a data descriptor and remains an open challenge for the community.

Changes made to the article Added a new paragraph, “Generalization across nominally identical platforms,” to the Usage Notes, addressing model transferability across structurally identical robot instances and the role this dataset’s calibration and parameter-identification procedures could play in characterizing a new platform’s residual physics.

Comment 3.2 *Revise figure captions to add brief explanatory notes for calibration sweeps and signal zero baselines for better standalone readability.*

Response to Comment 3.2 We agree and thank the reviewer. We revised the captions of Figure 3 and Figure 5 to explain the calibration sweeps and the signal zero baseline respectively, so each figure is interpretable on its own without cross-referencing the Methods text.

Changes made to the article The caption of Figure 3 was extended to state that the initial x - z and y - z excursions visible on the circular and Lissajous trajectories are calibration sweeps used to register the FBG sensor and OptiTrack frames. The caption of Figure 5 was extended to state that both sensors are tared immediately before recording, so each trace begins from a zero baseline.

Reviewer 4

As this is my first review of a dataset paper, I used an LLM to define criteria:

1. Data Description & Scope
2. Data Quality & Validation
3. Metadata & Documentation
4. Accessibility & Reusability (FAIR Principles)

5. *Technical Rigor & Reproducibility*
6. *Potential Impact & Use Cases*
7. *Ethical & Legal Considerations*

I will structure my review accordingly:

1. Data Description & Scope

The scope of the paper is a dataset on a tendon driven continuum robot with the aim to provide validation data for computational models predicting the behaviour of soft robots in the interaction with the environment. The introduction clearly states the purpose and nicely points out the value of dataset in other robotic disciplines highlighting the gap and need. A combination of static and dynamic scenarios and multi-modal data is a promising basis for developing methods for modelling of tendon-driven continuum robots.

The paper is very well written and comprehensive. From what I can judge, all relevant and even small details are described in the paper.

2. Data Quality & Validation

Raw and pre-processed data is provided. The Multi-modelity is very nice, as it allows details cross-validation between sensor modalities, which is performed already here in the paper. All of this looks very good.

3. Metadata & Documentation

The Dataset is well documented and in itself not super complicated. The Documentation in the paper is well done and should allow easy use of the dataset.

4. Accessibility & Reusability (FAIR Principles)

All data is accessible via figshare. I was able to download and inspect it and compare it to the paper description. It is well documented. A DOI is provided. Copyright: CC BY 4.0.

5. Technical Rigor & Reproducibility

Experimental setup and Data cross-validation are appropriate and technically sound. Reproducibility is given, as the robot components are commercially available or 3D printed and not too difficult to assemble.

6. Potential Impact & Use Cases

The dataset is relevant for a specific type of continuum robot, but it is argued that this is "the most widely studied paradigm in continuum robotics". For researchers interested in modelling this type of actuation, this dataset will be very valuable.

7. Ethical & Legal Considerations

None

One points for a minor revision:

Comment 4.1 *The type of robot is justified in the introduction briefly (commonly used continuum robot architecture) but I think this could be strengthened by a brief section on related robots and modelling attempts.*

Response to Comment 4.1 We agree and have expanded the Introduction with a brief discussion of recent modeling and control attempts specific to tendon-driven continuum robots, reinforcing why a shared dataset for this platform type is still missing.

Table 1: RMSE between OptiTrack and FBG-reconstructed disk position for the contact-motion recordings, with Lissajous fast as a non-contact reference. Values are absolute, in millimeters. Disk index k follows the base-to-tip convention ($k = 4$: tip).

Disk k	Recording	RMSE x [mm]	RMSE y [mm]	RMSE z [mm]
0	plane_y_contact_midway	0.099	0.078	0.054
	plane_y_poke_x	0.092	0.123	0.591
	plane_y_poke_y	0.045	0.110	0.111
	push_retract	0.124	0.117	0.089
	Lissajous fast (non-contact)	0.150	0.223	0.042
1	plane_y_contact_midway	1.386	2.849	0.750
	plane_y_poke_x	1.178	2.588	0.261
	plane_y_poke_y	0.852	2.634	0.748
	push_retract	1.450	2.316	1.013
	Lissajous fast (non-contact)	2.345	2.589	0.702
2	plane_y_contact_midway	2.738	5.751	1.058
	plane_y_poke_x	2.349	5.778	1.271
	plane_y_poke_y	1.838	5.863	1.099
	push_retract	2.640	4.662	1.009
	Lissajous fast (non-contact)	5.354	5.454	1.477
3	plane_y_contact_midway	4.598	7.984	2.489
	plane_y_poke_x	4.294	7.952	2.942
	plane_y_poke_y	3.914	8.056	2.559
	push_retract	3.770	7.090	0.912
	Lissajous fast (non-contact)	7.754	7.616	3.933
4	plane_y_contact_midway	7.086	9.443	4.617
	plane_y_poke_x	6.796	9.071	5.117
	plane_y_poke_y	6.680	9.256	4.694
	push_retract	4.488	9.629	1.071
	Lissajous fast (non-contact)	9.579	9.073	6.766

Changes made to the article Added a sentence to the Introduction citing recent data-driven modeling, control, and estimation work on tendon-driven continuum robots.

Comment 4.2 *I assume that in the force contact experiments, the bending sensors and optical MoCap will differ more as this may introduce additional subsegmental deformations? Did you check this? This would be interesting to mention in the technical validation.*

Response to Comment 4.2 We checked this directly. We compared the per-disk RMSE between the reconstructed FBGS shape and the corrected MoCap poses across the four contact recordings (contact_motion/plane_y_contact_midway, plane_y_poke_x, plane_y_poke_y, push_retract) against a representative non-contact dynamic recording (dynamic_motion/Lissajous_fast). Table ?? reports the absolute RMSE at every disk.

The RMSE during contact is at or below the level seen in the non-contact recording, at every disk and every axis. We find no evidence that contact introduces additional subsegmental deformation that the MoCap disks fail to capture.

We report absolute RMSE rather than RMSE normalized by motion range. During motion the intermediate disks move very little, so the same sub-millimetre discrepancy present throughout the dataset divides into a disproportionately large percentage. Absolute RMSE stays comparable across recordings regardless of motion range, which is why we report it here.

Changes made to the article To assess the reviewer concern, we extend the comparison of the four contact recordings (plane_y_contact_midway, plane_y_poke_x, plane_y_poke_y, push_retract), reporting the RMSE at every disk in Table ?? alongside Lissajous fast as a non-contact reference. The RMSE during contact is at or below the level observed in the non-contact reference at every disk and axis, indicating no measurable increase in OptiTrack-FBG discrepancy attributable to contact-induced local deformation.

Comment 4.3 *CC-BY 4.0 is specified on the figshare page, but not in the paper. This should be added for clarity. The code license is given (MIT)*

Response to Comment 4.3 We thank the reviewer for the careful reading, the dataset license has been mentioned in the article as well.

Comment 4.4 *2.2.4 first sentence is a bit unclear to me. "pose of selected..." Maybe: position and orientation of the robot discs (Ref to Fig 1)*

Response to Comment 4.4 We agree the original sentence was unclear. We have revised it to define "pose" explicitly as position and orientation, and to give concrete examples of the tracked objects rather than leaving "selected objects" unspecified.

Changes made to the article The first sentence of the "Rigid body poses" section was revised.

Comment 4.5 *If there are videos available from the recordings, I would encourage you to share them too. This could be very helpful.*

Response to Comment 4.5 We thank the reviewer for the suggestion. Unfortunately, video recordings of the data collection sessions were not made during data acquisition and are therefore not available to share.

Comment 4.6 LLM: Data Versioning & Updates *Is this dataset versioned? Will future updates (e.g., new trajectories, sensor calibrations) be tracked via a DOI or version tag? If not, consider adding a note about long-term maintenance.*

Response to Comment 4.6 We would like to reassure the reviewer on this point. The online figshare repository has its own versioning, so changes and new releases will be available under the same DOI, while previous versions can be still selected.

Comment 4.7 LLM: Use of the Dataset in the Paper *The paper uses the dataset to validate a model (GVS), but this is not a primary contribution. Consider adding a sentence in the 'Usage Notes' section: 'This dataset has already been used to validate a forward dynamics model (GVS) in [citation], demonstrating its utility for benchmarking.'*

Response to Comment 4.7 We thank the reviewer for this suggestion. The suggested sentence assumes an external, previously published use of the dataset; here, the analogous check is instead reported directly in this manuscript's Technical Validation section. Per the Editors' guidance that Data Descriptors should not include additional results or analysis, we have added a brief pointer in Usage Notes to where tendon tension and base wrench measurements are used together with a dynamics model, without restating any outcome.

Changes made to the article Added one sentence to Usage Notes pointing to the dynamics-model-based check reported in Technical Validation.

Reviewer 5

general comment *The work presents a dataset for a single-segment soft robot, built from commercially available components. The dataset contains readings from six sensor modalities in quasi-static and dynamic motions as well as contact situation. Cross-validation was undertaken for consistency assessment. Overall, the work addresses a general lack of comparable, transferable, standardized and available experimental data in the field of soft robotics. The scientific contribution and relevance of the work is well motivated and aims at filling an existing research gap in the field of soft robotics. However, the paper and the dataset show major weaknesses and inconsistencies, which should be revised before the paper can be published. Details below.*

Response to the general comment

Major comments (paper)

Comment 5.1 *The introduction to the work is thoroughly written and also effectively compares the state of the art in soft robotics well with other data-intensive research fields such as autonomous driving and robotics in general. However, the introduction lacks comparisons to datasets in the same field, which should be added/referred to, e.g., [1, 2, 3, 4, 5].*

Response to Comment 5.1 We agree and thank the reviewer for the suggested references. We have added them to the Introduction alongside a brief discussion of recent data-driven modeling and control attempts for tendon-driven continuum robots, each validated on its own custom platform – reinforcing the motivation for a shared, reusable dataset.

Changes made to the article Added the five suggested references ([1, 2, 3, 4, 5]) to the Introduction.

Comment 5.2 *While the experimental setup is described in detail, the figures (especially Fig. 2) should be much clearer:*

- *Inconsistent writing (lowercase / uppercase). Example: Contact probe vs. Reflective Marker.*
- *Coordinate frames are barely visible, especially when printed on paper in greyscale. Increase contrast.*
- *Dashed lines and circles are not aligned well. Correct alignment.*

Response to Comment 5.2 We thank the reviewer for the input. To improve the clarity of the manuscript Figure 1 and 2 have been updated using consistent font and writing convention. The dashed lines and dot have been aligned to the drawing.

Changes made to the article Figure 1 and 2 were changes as mentioned above.

Comment 5.3 *Multiple qualitative terms such as “rigidly connected” and “geometrically stable” are used. Such vague terms are too unprecise in an experimental paper about soft robotics, which heavily relies on a sufficiently rigid mounting structure. The authors should provide quantitative measurements such as static directional stiffness values or structural eigenfrequencies to quantify that the mounting structure is sufficiently stiff for the experiments.*

Response to Comment 5.3 We agree that “rigid and geometrically stable” was too vague. We quantified the frame-to-base mounting stiffness with a dedicated bench test: while logging the base ATI force/torque sensor, we applied a series of manual pushes to the robot base and tracked the resulting displacement between markers on the base disk and markers on the support frame with the OptiTrack system. Across five pushes spanning different force directions (25 N to 48 N), the measured stiffness ranged from 56 N mm^{-1} to 316 N mm^{-1} depending on direction, confirming the mount is anisotropic. The weakest direction (56 N mm^{-1} , along the robot’s x -axis) is still more than sufficient for the loads the robot itself generates during operation: across all recorded trajectories, the in-plane forces and torques transmitted through this connection peak at 0.4 N to 0.8 N and 0.15 N m respectively, which at the measured stiffness correspond to sub-0.01 mm deflections, well below the resolution of any sensor in the dataset. The largest base reaction the robot produces is along its z -axis (tendon-pretension dominated, up to 40 N), but this load is balanced by the motor pull on the tendon, which is directly distributed on the vertical profiles. We have added these quantitative values to the manuscript.

Changes made in the manuscript Replaced “forming a rigid and geometrically stable structure” and the following sentence about minimizing structural compliance with a brief description and a forward pointer to a new Technical Validation subsection. In Technical Validation the new subsection “Mounting stiffness” presents the measured directional mounting stiffness (56 N mm^{-1} to 316 N mm^{-1} depending on direction) from the dedicated bench test, and compares it against the actual base loads generated by the robot during operation to show the resulting deflections are negligible.

Comment 5.4 *The contact-force validation remains largely qualitative. Terms such as “close agreement” and “good agreement” should be supported by quantitative metrics such as RMSE, mean bias, maximum absolute error, and preferably correlation values.*

Response to Comment 5.4 We have replaced the qualitative “close agreement”/“good agreement” language with quantitative metrics (RMSE, mean bias, maximum absolute error, and Pearson correlation r), restricted to samples where the Resense wand is actually in contact. To exclude the retracted/no-contact portions of each recording, contact is identified geometrically, from the mocap-tracked distance between the wand and the surface it presses on.

Changes made to the article Added a table reporting RMSE, mean bias, maximum absolute error, and correlation for F_x and F_y , contact samples only, for both the `touch_base` and `push_retract` recordings.

Comment 5.5 *Inconsistency in Tab. 1: The table does not only contain geometrical parameters. Correct the description of the table.*

Response to Comment 5.5 We agree. The table caption stated “Geometrical parameters,” but the table also reports material properties (tendon material, backbone Young’s modulus) and mass values for the backbone, disks, and marker sets.

Changes made to the article The caption of Table 1 was corrected to “Geometrical, material, and mass parameters of the experimental platform.”

Comment 5.6 *Inconsistency in Tab. 1: The physical unit of weight is newtons (N), not grams (g). Refer to mass, not weight, when using grams as the physical unit. Correct units.*

Response to Comment 5.6 We thank the reviewer for the careful reading.

Changes made to the article Since the quantity was correctly expressed in grams, the label as been changed from “Weight” to “Mass”.

Comment 5.7 *Inconsistency with kinematic definitions: g_{fix} is introduced as a rotation matrix (orientation), but g_{wand} is introduced as a pose. In robotics, a pose is position and orientation.*

Response to Comment 5.7 We agree with the reviewer that a pose is composed by a position and orientation components. We would like to highlight that this is the case for g_{fix} introduced in section *Data collection accessory*.

Changes made to the article We clarified in *Data collection accessory* the composition of the pose with explicit reference to its position and orientation parts.

Comment 5.8 *The paper currently lacks a clear, dedicated conclusion and summary section. The authors should add a concluding section that summarizes the impact of the dataset and outlines potential future use cases.*

Response to Comment 5.8 We agree in principle, but a dedicated Conclusion is not standard for a Scientific Data descriptor. The Editors’ own guidance for this submission (see Response to Editor comments, above) states that Data Descriptors should not include significant results, conclusions, discussion or data analyses. We have therefore not added a Conclusion section.

Changes made to the article No change was made to the article.

Comment 5.9 *The paper currently lacks an application to a baseline model for at least one of the use cases that are mentioned in the introduction. Therefore, the paper does not demonstrate by itself that the dataset is useful for any of these applications. Neither does it demonstrate that the authors have successfully used their dataset. This could have helped with the inconsistencies of the dataset (see further below).*

Response to Comment 5.9 We understand the reviewer request. However, even if we would like to provide an example of usage of our dataset, we are restricted by the Data Descriptor guidelines, preventing the presentation of any result or conclusion. We would like to note that an usage example is already present: Technical Validation includes a dynamics-model-based consistency check using tendon tension and base wrench measurements. It’s usage was justified by the need of having a physical model relating two heterogeneous force measurement.

Changes made to the article We have extended the cross-validation sentence in Background & Summary to name the dynamics-model-based consistency check reported in Technical Validation.

Comment 5.10 *The GVS model used for Figure 6 and Table 7 is not included in the released code. Since this model forms part of the technical validation, the implementation and all parameters required to reproduce the reported results should be released.*

Response to Comment 5.10 We understand the reviewer request. The simulation code was retained as was not considered part of the dataset acquisition or postprocessing pipeline. However, being it part of the technical validation it should indeed be present in the repository.

Changes made to the article We mentioned the presence of the simulation code in Section Consistency of tendon tension measurements while addressed its usage in Usage Notes.

Minor comments (paper)

Comment 5.11 *The word “minimize” should only be used when referring specifically to the results of an optimization routine. Consider using “reduce” or “eliminate” to avoid confusion. This occurs multiple times in the paper and should be corrected.*

Response to Comment 5.11 We thank the reviewer for the terminology clarification.

Changes made to the article The word “minimize” has been replaced by “reduce”.

Comment 5.12 *Sec. 2.2: 3D printed pulleys appear to be an unnecessarily low-cost option for routing cables. Have the authors checked for roundness (part may have a seam from printing) and eccentricity after pressing the bearing in? It remains unclear why 3D printed pulleys were used at all, as machined parts are widely available standard components.*

Response to Comment 5.12 The pulley diameter had to match the offset between the actuator spool and the tendon routing hole in the robot exactly, a dimension that off-the-shelf pulleys do not provide; we therefore opted for custom 3D-printed pulleys. We confirm the pulleys are mounted on a bearing, pressed into the printed part’s bore. The bore-to-bearing clearance is small enough that the press fit does not measurably deform the printed pulley, and roundness/concentricity was verified on a cylindrical test bench prior to assembly; all pulleys were printed on a Bambu Lab H2D, chosen for its dimensional accuracy and repeatability.

Changes made to the article In Methods, the Tendon Tension subsection has been modified to clarify the necessity of 3D printed pulleys.

Comment 5.13 *While mathematically identical for orthogonal matrices, it is rather common in robotics to use the transpose of a rotation matrix instead of its inverse (see inversion of g_{base} on page 16). The authors may want to use the transpose for stylistic consistency.*

Response to Comment 5.13 We agree with the reviewer that for orthogonal matrices the transpose and inverse are equivalent operations. However, as mentioned in 5.7, g_{fix} and g_{base} are homogeneous transformations, thus requiring the inverse operation to compute a relative transformation.

Changes made to the article No changes were made to the article.

Comment 5.14 *Some parts of the paper read more like a technical documentation rather than a scientific paper. Overall, the writing could be improved before publication so that the text reads more fluently.*

Response to Comment 5.14 We thank the reviewer for the feedback. After careful reading, we spotted several section that presented the technical details without proper introduction.

Changes made to the article In Sections Data Collection (Postprocessing and Data collection interfaces) and Methods (Session preparation), several paragraph have been introduced to guide the reader through the technical explanations.

Comment 5.15 *Data availability and code availability refer to the same internet source, so it could be included in a single section rather than two short sections.*

Response to Comment 5.15 We agree with the reviewer that the standard Data Descriptors formatting is not necessary in this instance **period**

Changes made to the article The two sections have been merged **period**

Comment 5.16 *Table 3 states that the FBG shape contains 502 points but uses the index range 0–502, which would correspond to 503 points.*

Response to Comment 5.16 We thank the reviewer for the careful reading.

Changes made to the article To remain consistent with the MATLAB indexing, we indicate the points range 1–502.

Major comments (dataset)

Comment 5.17 *The documented column order of `mocap_frames.csv` appears to be incorrect. Table 3 states $[x, y, z, \text{roll}, \text{pitch}, \text{yaw}]$, whereas both the released processing code (`data_optitrack.m`, lines 143–149; `process_data.m`, lines 1049–1066) and inspection of the released CSV indicate $[\text{roll}, \text{pitch}, \text{yaw}, x, y, z]$. This can directly lead to incorrect interpretation of the dataset and should be corrected before publication.*

Response to Comment 5.17 We agree and thank the reviewer for the precise cross-check. We confirm the released code and CSV files are correct; only Table 3’s documented column order was wrong. It has been corrected to list orientation (roll, pitch, yaw) before position (x, y, z) , matching `data_optitrack.m`, `process_data.m` and the released CSVs.

Changes made to the article Table 3’s `mocap_frames.csv` row was corrected to list disk orientation before disk position, matching the actual column order in the released files.

Comment 5.18 *A Python logic error is present in `command_motors.py` (name changed during revision from `read4MotorCircle`): `if motion_pattern == "circle" or "Lissajous":`. This condition is always true because the non-empty string literal "Lissajous" evaluates to true in a Boolean context in Python. As a consequence, the angular tolerance is apparently set to 12° for all motion types rather than the intended 0.5° default. The authors should clarify whether this code version was used for the published experiments and assess its impact on the recorded trajectories.*

Response to Comment 5.18 This is indeed an overlooked logic error, but it has no impact on the recorded trajectories. The purpose of this line was only to make the motor come sufficiently close to the current waypoint before moving to the next waypoint, so that the tip trajectory looks closer to a circle or Lissajous. However, since we are using only a few waypoints and the trajectories do not have to be a perfect circle, this logic error has no impact on the trajectories themselves and the purpose they mean to serve. We have nonetheless corrected both instances of the logic error (the tolerance check at line 57 and the phase-transition speed limit at line 299) in the released code, so the repository reflects the intended behaviour going forward; this has no effect on the released data.

Changes made to the code The logic error was corrected in the released acquisition code (`command_motor.py`, name updated during the revision process).

Comment 5.19 *The filtering procedure may be seriously affected by irregular timestamps. `process_data.m` determines the sampling frequency using $F_s = 1/\text{median}(\text{diff}(t))$ and applies a standard Butterworth filter before uniform resampling. This assumes approximately equidistant samples. The inspected ATI raw recording exhibits strongly burst-like host timestamps despite an average data rate of approximately 200 Hz, meaning that the median timestamp interval may yield a grossly incorrect filter sampling frequency. The authors should verify whether the released processed signals were affected and, if necessary, regenerate them using a regular time grid before filtering.*

Response to Comment 5.19 We confirm the diagnosis. `butter_filtfilt` (the shared filtering utility used throughout `process_data.m`) estimated each sensor’s sampling rate as $f_s = 1/\text{median}(\text{diff}(t))$, which is only reliable when timestamps are close to evenly spaced. We audited the raw host timestamps of every sensor across all released recordings, computing the mean and median inter-sample interval and their ratio (Table 2). The ATI force/torque stream (mini40) is strongly bursty: mean/median ratio of 26.3, driven by a block-read acquisition pattern (most gaps under 1 ms, the remainder near 15 ms), giving an estimated f_s about 26 times too high and an effective filter cutoff far below the intended 15 Hz. The motor encoder stream shows a smaller but real bias (ratio 1.41). OptiTrack, FBG, Mark-10 and HEX 12 timestamps are close to uniform. To address this, `process_data.m` now estimates each sensor’s f_s from the mean inter-sample interval after resampling its timestamps onto a uniform grid, and all processed files have been regenerated with the corrected filter.

Table 2: Mean and median inter-sample interval of each sensor’s raw host timestamps, pooled across all released recordings.

Sensor	Mean interval [ms]	Median interval [ms]	Mean/Median
ATI mini40	5.00	0.19	26.3
Motor encoders	2.82	2.00	1.41
FBG sensor	10.01	9.76	1.03
OptiTrack	8.33	8.33	1.00
Mark-10	2.85	2.80	1.02
HEX 12	9.84	10.00	0.98

Changes made to the code In `process_data.m`, the shared filtering utility `butter_filtfilt` has been modified to estimate each sensor’s sampling rate from the mean inter-sample interval and to re-sample its timestamps onto a uniform grid spanning the same time range and sample count.

Changes made to the article A clarifying sentence describing this was added to the “Low-pass filtering” paragraph in Methods. All processed files were regenerated with the corrected filter and are included in this revision.

Comment 5.20 *Contact-data processing is inconsistent between the manuscript and the released code. The manuscript specifies a 15 Hz low-pass filter, whereas `compare_ft_sensors.m` uses 30 Hz. Processed files are described as header-less, while the contact script uses `writetable`. File names also differ (`tendon_tensions.csv` vs. `cable_tensions.csv`, `contact_wrench.csv` vs. `wrench_wand.csv`), and no clear generator for the documented `wand_pose.csv` was identified.*

Response to Comment 5.20 We agree, and thank the reviewer for the detailed cross-check. The 15 Hz filter and header-less convention described in the manuscript are correct and match `process_data.m`, the sole generator of the canonical `processed/` files, including for the contact subset. The 30 Hz filter and `writetable` headers the reviewer found belonged to a separate, now-superseded script (named `compare_ft_sensors.m`); the postprocessing pipeline has since been streamlined so that `process_data.m` is the only script that filters and writes the released `processed/` files. Contact-force validation is now performed by a dedicated script, which reads the processed data and applies one additional low-pass filter (2 Hz), specific to this comparison: the applied contact force is small, so residual noise passed by the 15 Hz filter would otherwise inflate the reported RMSE. Furthermore, we corrected the pipeline so that the wand pose, for recordings with the Resense sensor, is released separately in `wand_pose.csv`. We also corrected the one remaining filename mismatch: the released name is `wrench_wand.csv`, not `contact_wrench.csv` as listed.

Changes made to the code Contact-force validation is now performed by the dedicated script, namely `tests/compare_ati_resense_wrench.m`. The postprocessing pipeline has been streamlined so that `process_data.m` is the sole generator of the released `processed/` files, and now unconditionally writes `mocap_frames.csv` with the robot’s 5 disks only, releasing the Resense wand’s pose separately as `wand_pose.csv`. Output files names in `process_data.m` have been corrected for the wand pose and contact wrench.

Changes made to the article Clarification about the new additional filter was added in Section Technical Validation (Consistency of contact measurements). In Section Data Records (File naming and outputs), we clarified that the Subset 3 contains the pose of the wand and the contact wrench.

Comment 5.21 *The main acquisition script does not reproduce the contact subset. `runAllDataCollection.bat` starts FBG, ATI, OptiTrack, four Mark-10 processes, motors, and an undocumented gyroscope, but does not start the Resense/HEX12 acquisition process required for the contact experiments.*

Response to Comment 5.21 We thank the reviewer for the careful reading. The reviewer is correct: `runAllDataCollection.bat` was not updated to the version actually used on the acquisition machine: it contained an obsolete line starting a gyroscope process that was used only during early prototyping and was not part of the sensor suite for the published contact experiments. The call to the Resense/HEX12 acquisition process has since been added to the script on the acquisition machine, but this change had not yet been synced to the repository at the time of the original submission.

Changes made to the code The acquisition code `runAllDataCollection.bat` has been updated.

Comment 5.22 *The released code is not yet a fully reproducible standalone processing pipeline. Important parameters and paths are hard-coded in `process_data.m`, no complete top-level README or dependency file is provided, and several required software/hardware dependencies are not documented. A clean, parameterized Raw $\rightarrow$ Processed workflow should be provided. The manuscript and the released processing code describe different OptiTrack calibration procedures. Section 2.6.3 states that the per-disk marker offsets are determined once from `references/straight_config`, whereas `process_data.m` (lines 141–195) recalculates these offsets independently from the first 3 s of every recording. The authors should clarify which procedure was actually used to generate the released processed data.*

Response to Comment 5.22 We thank the reviewer for these three points, which we address in turn.

OptiTrack calibration. The per-disk static-misalignment correction is now computed once, by averaging each disk’s pose over the entire `references/straight_config/` recording (`utils/compute_mocap_correction.m`), and saved to a single shared file (`data/postprocess_calibration/mocap_correction.csv`) that every recording’s processing run simply loads.

Hardcoded paths. The released code and data now share the same relative layout distributed on figshare (`code/` and `data/` as siblings), and every path used by the post-processing pipeline is relative to that layout. All editable settings, including the path to the experiment being processed, are grouped in a single `PATHS/SETTINGS` block at the top of `process_data.m` (the only file that needs to be run). The pipeline

runs unmodified on any machine once the repository is downloaded, with nothing machine-specific to edit elsewhere.

README, dependencies, and workflow documentation. We have added a `README.md` and a `DEPENDENCIES.md` to the root of the code archive. The `README` documents the repository layout, the expected raw-data layout, and the full raw-to-processed workflow step by step, including which two calibration files are computed once on first run and reused thereafter. `DEPENDENCIES.md` lists the software (Python packages and MATLAB/toolbox requirements) and hardware required for both the acquisition and post-processing stages.

Changes made to the article We have added one sentence to the "Coordinate frame alignment" paragraph (Section Data Collection) explicitly distinguishing the once-per-dataset per-disk correction from the once-per-recording base-frame definition. We have added one sentence to the "Software requirements" paragraph (Usage Notes) stating that all pipeline paths are relative and centralized in a single settings block, so the code is machine-agnostic. We have completed the previously blank Code Availability statement, describing the code's license and the new `README.md` and `DEPENDENCIES.md` now included with it, along with the acquisition and post-processing software requirements.

Response to Comment 5.22 We thank the reviewer for these three points, which we address in turn.

OptiTrack calibration. The per-disk static-misalignment correction is now computed once, by averaging each disk's pose over the entire `references/straight_config/` recording, and saved to a single shared file that every recording's processing run simply loads. The released processed data has been regenerated using this corrected, once-only procedure.

Hardcoded paths. The released code and data now share the same relative layout distributed on figshare (`code/` and `data/` as siblings), and every path used by the post-processing pipeline is relative to that layout. All editable settings, including the path to the experiment being processed, are grouped in a single `PATHS/SETTINGS` block at the top of `process_data.m` (the only file that needs to be run). The pipeline runs unmodified on any machine once the repository is downloaded, with nothing machine-specific to edit elsewhere.

README, dependencies, and workflow documentation. At the root of the code archive, we have added a `README.md` and a `DEPENDENCIES.md`. The `README` documents the repository layout, the expected raw-data layout, and the full raw-to-processed workflow step by step, including which two calibration files are computed once on first run and reused thereafter. `DEPENDENCIES.md` lists the software (Python packages and MATLAB/toolbox requirements) and hardware required for both the acquisition and post-processing stages.

Changes made to the code *OptiTrack calibration.* Added `utils/compute_mocap_correction.m`, which computes the per-disk static-misalignment correction once from the `straight_config/` recording and writes it to `data/postprocess_calibration/mocap_correction.csv`; `process_data.m` now loads this file instead of recalculating the offsets per recording. All processed files were regenerated with this corrected procedure and are included in this revision.

Hardcoded paths. `process_data.m` restructured so all paths are relative to the `code//data/` layout distributed on figshare, with every editable setting collected in a single `PATHS/SETTINGS` block at the top of the file.

Documentation. Added `README.md` (repository layout, raw-data layout, full raw-to-processed workflow, which calibration files are cached) and `DEPENDENCIES.md` (software and hardware requirements for acquisition and post-processing) to the root of the code archive.

Changes made to the article Added one sentence to the "Coordinate frame alignment" paragraph (Section Data Collection) distinguishing the once-per-dataset per-disk correction from the once-per-recording base-frame definition. Added one sentence to the "Software requirements" paragraph (Usage Notes) stating that pipeline paths are relative and centralized in a single settings block, and mentioned the new `README.md` and `DEPENDENCIES.md`, and the acquisition/post-processing software requirements. Completed the Code Availability statement, describing the code's license.

Comment 5.23 *The description of the circle and Lissajous trajectories does not fully match the acquisition code. The manuscript describes continuous sinusoidal commands, whereas `command_motors.py` implements the circle using discrete target configurations and the Lissajous trajectory using 16 sampled waypoints that are sequentially executed by position control. The actual actuation protocol should be described accurately.*

Response to Comment 5.23 We thank the reviewer for this observation and confirm the description was imprecise. Neither trajectory uses continuous sinusoidal commands: both are executed as a sequence of discrete target configurations under closed-loop position control. The `Lissajous` waypoints are generated in code from the parametric equations $x(t) = R\sin(at + \delta)$, $y(t) = R\sin(bt)$ (sixteen waypoints per cycle, with $a = 2$, $b = 1$, $\delta = 0$, $R = 180^\circ$). The `circle` trajectory, by contrast, uses eight fixed, hand-tuned waypoints that approximate the same nominal circle ($a = b = 1$, $\delta = \pi/2$), with small adjustments introduced to compensate tendons slack. Both trajectories apply a 15% reduction to the commanded offset of the tightening tendon in each antagonist pair for the same reason. We have revised the manuscript to describe this accurately, distinguishing the formula-generated `Lissajous` waypoints from the hand-tuned `circle` waypoints, and to note the tendon-slack compensation and its consequence for users of the commanded-angle data.

Changes made to the article The description of the `circle` and `Lissajous` trajectories in the Dynamic motion subset (Data Records) has been revised to state that both are executed as discrete target-configuration sequences under position control, to give the parametric equations and parameter values underlying the `Lissajous` waypoints, and to clarify that the `circle` waypoints are hand-tuned rather than formula-generated.

Comment 5.24 *The quasi-static acquisition protocol cannot be reproduced from the released acquisition code. The manuscript describes small incremental sweeps through near-equilibrium configurations, while the available code mainly contains two extreme target configurations and no explicit equilibrium criterion. Either the actual acquisition script should be released or the Methods section should be revised accordingly.*

Response to Comment 5.24 We would like to clarify this point. The released acquisition script (`command_motors.py`) is the one actually used for the quasi-static recordings; the confusion arises because, for this particular motion pattern, the target amplitude (`desired_degree`) is a single hardcoded value rather than a command-line argument, and was edited by hand between recordings. As released, the script therefore only shows the value used for the last of the 16 quasi-static recordings (180°). Each recording drives one antagonist tendon pair back and forth between two extreme target configurations at 0.3rad/s, advancing between them once the motor reaches the target within a position tolerance; the resulting slow, densely sampled traversal between the two extremes is what produces the sequence of near-equilibrium configurations reported in the manuscript. On this point, with near-equilibrium configuration we intend that the robot moves sufficiently slowly so that, once the target angle is reached by the actuators, it presents negligible inertia effect, allowing the quasi static assumption to hold.

Changes made to the article A sentence has been added in Section Recording protocol to clarify the manual input used to generate the trajectories. In Quasi-static subset paragraph of Recording Protocol, a sentence has been added to clarify the near-equilibrium configurations.

Comment 5.25 *The temporal synchronization discussion appears too strong. Although all sensor processes use the same host clock, their timestamps correspond to different events such as serial-response completion, DAQ block reads, sequential CAN queries, network reception, or processed FBG-frame reception. A common OS clock therefore does not necessarily imply a common physical measurement time.*

Response to Comment 5.25 We understand the reviewer’s concern: a shared OS clock rules out drift between streams, but it does not establish that all timestamps mark the same stage of measurement.

Changes made to the article We have revised Synchronization in Section Data Collection and Temporal synchronization in Section Technical Validation to state this explicitly and to remove the claim that the motor, ATI and Mark-10 channels carry no additional latency beyond OS jitter.

Comment 5.26 *The manuscript states that cross-correlation was omitted for tendon tension and base force/torque signals, while the released synchronization code nevertheless computes motor-tension and motor-ATI correlations. The authors should clarify whether these were exploratory analyses only and align the released validation code with the description in the paper.*

Response to Comment 5.26 We thank the reviewer for catching this inconsistency. The motor-tendon-tension and motor-ATI cross-correlations present in the released synchronization scripts were leftover exploratory diagnostics from an earlier stage of the analysis; they were never used to support any claim in the manuscript. As the reviewer’s comment implies, they should not have been presented as part of the validation code

Changes made to the code We have removed this section of the temporal synchronization from the released files. The now updated `check_temporal_sync.m` and `compute_sensors_delay.m` compute only the motor-Mocap, motor-FBG, and Mocap-FBG cross-correlations.

Changes made to the article A clause has been added to the Temporal synchronization paragraph (Technical Validation) and the reported values have been modified to comply with the result of the new cross-correlation script.

Comment 5.27 *The FBG-OptiTrack comparison should be described more carefully as a cross-modal consistency check after frame registration. Since OptiTrack data are used to align the FBG frame before the two modalities are compared, this is not a completely independent absolute validation.*

Response to Comment 5.27 We understand the concern and have reworded the relevant paragraphs. The FBG sensor’s bending plane is registered to the OptiTrack frame via the per-recording planar calibration sweeps before the tip trajectories are compared, so this is not an independent absolute-pose validation. The comparison is now described as a cross-modal shape-consistency check after frame registration.

Changes made to the article The introductory and concluding sentences of Section Kinematic cross-validation have been revised to state that the FBG frame is registered to OptiTrack via the calibration sweeps prior to comparison, and that the reported RMSE reflects cross-modal shape consistency after registration, not independent absolute validation.

Minor comments (dataset)

Comment 5.28 *The OptiTrack validity flags (`disk_k_is_valid`) are loaded in `data_optitrack.m` but apparently not used during processing, and the validity information is not retained in the processed data. The handling of occluded/invalid frames should be documented and preferably retained as a validity mask.*

Response to Comment 5.28 Concerning the validity flags, we relied on their positive values when processing our own data locally and had verified that all frames are valid. We understand the reviewer’s concern, and to address it explicitly we wrote a dedicated test that checks this flag across every released recording and reports, per disk and per recording, how many samples are valid versus invalid. Run against the full released dataset it checked 30 recordings, 154 disk-recording pairs, and 843 028 individual disk-frame entries, and found every one valid.

Changes made to the code A new script `check_optitrack_frame_validity.m` (in the tests folder) has been added to check the validity flag for all recordings.

Changes made to the article We have added a sentence to the manuscript stating this result explicitly in Section Comparison of recorded shape with FBG sensor and Motion Capture.

Comment 5.29 *The FBG delay value is missing from the Methods text (“a fixed offset of is subtracted”), while the code uses approximately 13.5 ms. The numerical value and sign convention should be stated explicitly.*

Response to Comment 5.29 We agree, and note the value has changed since the code version the reviewer inspected. At the time of the original submission, `process_data.m` applied FBG timestamp correction near 13.5 ms. As part of addressing Comment 5.30, this was replaced by the empirically measured value from `compute_sensors_delay.m`, 12.25 ms (the mean mocap-FBG lag over the four fast dynamic trajectories), which is what the released code now applies and what Technical Validation reports. The correction is subtracted from the FBG timestamp ($t_{FBG} \leftarrow t_{FBG} - 12.25 \text{ ms}$), consistent with the FBG pipeline’s processing latency relative to Mocap.

Changes made to the article Added the numerical value (12.25 ms) and its sign convention explicitly to the Methods section, keeping the reference to the Temporal synchronization section for its computation.

Comment 5.30 *Delay estimation and delay correction are mixed in `process_data.m`: the FBG timestamps are corrected before the synchronization check is executed. The lag estimation should preferably be performed on the uncorrected signals and separated from the subsequent correction step.*

Response to Comment 5.30 We understand the reviewer concern. In the original version of the code, the `process_data.m` was used to characterize the delay which was then corrected in the same code. This could lead to interpret the reported synchronization check as circular. To address this issue, we separated the delay estimation and correction using a dedicated script (`compute_sensors_delay.m`) for the former.

Changes made to the code In the new code release, we added `compute_sensors_delay.m`, which estimates the FBG pipeline delay independently of the timestamp correction, and `process_data.m` now cites its measured values in a comment justifying the correction constant it uses.

Comment 5.31 *The manuscript refers to `std::chrono::high_resolution_clock` for FBG timestamps, whereas the released C++ code stores timestamps using `std::chrono::system_clock`.*

Response to Comment 5.31 The manuscript incorrectly named the source of the FBG timestamp. In the released C++ client, `high_resolution_clock` is used only to bound the acquisition loop’s runtime; each frame’s actual timestamp is taken from `std::chrono::system_clock`, which is epoch-anchored and consistent with the wall-clock timestamps used by the other acquisition processes. We have corrected the manuscript text accordingly.

Changes made to the article The Data collection interfaces section was corrected to reference `std::chrono::system_clock` instead of `std::chrono::high_resolution_clock`.

Comment 5.32 *All four sequentially queried motor states are stored using a single timestamp. Either individual timestamps should be used or the total four-motor readout latency should be quantified.*

Response to Comment 5.32 We thank the reviewer for this suggestion. We measured the four-motor sequential CAN readout latency directly, using the same polling call and motor order used during acquisition, over 17,851 polling cycles collected across a 30-second session. The total four-motor readout latency averages 1.68 ms (std 0.09 ms, max 6.85 ms), i.e. each individual motor query takes approximately 0.42 ms and the four are queried back-to-back on the shared CAN bus.

Changes made to the article The data-acquisition description (Methods) now addresses that the four motor states sharing a single timestamp are read sequentially over CAN, while the Temporal synchronization sections reports the measured total four-motor readout latency.

Comment 5.33 *The ATI timestamp is assigned after reading and averaging a five-sample DAQ block. The timestamp therefore represents approximately the end rather than the center of the measurement block.*

Response to Comment 5.33 We agree and thank the reviewer for the precise diagnosis. The mini 40 timestamps were assigned when the blocking read returned – i.e. at the end of that window rather than its center. We have corrected this in the released processing pipeline: ATI timestamps are now shifted by -2.5 ms (half the averaging window) so that they mark the window’s temporal center.

Changes made to the article In Section Methods the mini 40 paragraph has been updated to state the time-stamping convention explicitly.

Comment 5.34 *The manuscript specifies a 2 s pre-motion period, while `command_motors.py` uses 3 s. This should be made consistent.*

Response to Comment 5.34 We thank the reviewer for pointing this out. We checked this directly against the released raw data: for every recording in the quasi-static, dynamic-motion, and contact subsets, we compared the timestamp of the first Mark-10 sample (logged immediately on process start) against the first motor sample (logged only after the pre-motion wait), since both processes run on the same acquisition machine and share the same wall clock. The resulting delay is consistent at approximately 3.5 s across all recordings checked. This confirms that the released acquisition script, which sets the delay to 3 s, is the one that was actually used throughout data collection, and that the manuscript’s stated value of 2 s was incorrect. We have corrected the manuscript accordingly.

Changes made to the article The pre-motion delay stated in Data collection interfaces has been corrected from 2 s to 3 s, matching the value used during data collection.

Comment 5.35 *The common 100 Hz resampling interval is based primarily on the motor recording duration rather than the common overlap of all sensor streams. The authors should verify that this does not introduce undefined or NaN edge samples.*

Response to Comment 5.35 We thank the reviewer for raising this. Checking every released recording directly, we confirmed this does introduce NaN edge samples: 15 of the 30 recordings have trailing NaNs in `tendon_tensions.csv`. In those recordings the corresponding Mark-10 gauge stopped logging a few tens of milliseconds before the motor did, and the resampling window was previously bounded only by the motor’s duration. No other output file is affected. We corrected `process_data.m` to bound the resampling window to the common overlap of all sensor streams (motor, all four tendon gauges, ATI, Mocap, FBGS): the window start is the latest of each stream’s first timestamp, and the window end is the earliest of each stream’s last timestamp. We regenerated the affected recordings with this corrected window.

Changes made to the code We corrected the resampling window in `process_data.m` to span the common overlap of all sensor streams — starting at the latest of each stream’s first timestamp and ending at the earliest of each stream’s last timestamp — rather than the motor recording alone, and regenerated and re-uploaded the affected recordings in the data release.

Comment 5.36 *Processed signals are generated using zero-phase `filtfilt`, which is acausal. This is appropriate for offline analysis but should be explicitly stated in the Usage Notes because the dataset is also proposed for online/state-estimation applications.*

Response to Comment 5.36 We have added an explicit statement to the Usage Notes that `filtfilt` is acausal and cannot be reproduced in a causal, online or real-time setting, and that users requiring causal filtering should apply their own causal filter instead.

Changes made to the article Usage Notes was revised to state this explicitly.

Comment 5.37 *The reported Mark-10 noise values should be checked. `analyze_reference_noise.m` appears to calculate the standard deviation from the raw pulley reaction force, while normal processing divides these values by two to obtain tendon tension.*

Response to Comment 5.37 We thank the reviewer for the careful reading. The script indeed computed the standard deviation from the raw pulley reaction force rather than the tendon tension; we corrected it to divide by 2, consistent with the convention used everywhere else in the pipeline (`process_data.m`). For this reference recording (`references/released_config`, robot at rest, tendons released), the correction does not change the reported value: with the tendons unloaded, all four Mark-10 gauges report an identically zero reading throughout the recording, so the computed standard deviation is 0 N regardless of the scaling factor. The 0.02 N value in Table 4 is therefore not this recording’s measured standard deviation, but the M3-5 gauge’s manufacturer-specified resolution.

Changes made to the article We added a footnote to Table 4 clarifying that the Mark-10 values are the manufacturer-specified resolution rather than a measured standard deviation, since the sensor reads identically zero at zero load.

Comment 5.38 *The “Euclidean distance” used for contact-position validation is calculated only in the xy plane in `compare_ft_sensors.m`. It should either be referred to as planar distance or calculated in full 3-D.*

Response to Comment 5.38 The released `compare_ft_sensors.m` computed this distance using only the x and y components (`disk_xy`, `sensor_xy`), even though the z component was already available in the same script. This choice was driven by the experimental condition where only x and y displacements were present between the rigid bodies. However, we have corrected the code to compute the full 3-D Euclidean distance between the tracked wand position and the robot tip disk centre. Recomputing this metric over the full `push_retract` recording yields the same result reported in the manuscript (mean ≈ 38 mm, negligible standard deviation), confirming the validation conclusion is unaffected.

Changes made to the code The processing code (`compare_ati_resense_wrench.m`) was corrected to compute the Euclidean distance using all three position components (x , y , z). No change to the manuscript text or reported value was necessary, since the recomputed distance matches the figure already reported.

Comment 5.39 *The FBG files referred to as “raw” already contain proprietary Shape CORE reconstructed curvature, bending angle, and 3-D shape rather than raw optical FBG measurements. The terminology should be clarified.*

Response to Comment 5.39 We agree and thank the reviewer for flagging this. The Data Records section now states explicitly that the FBG “raw” file (`dataFBGS.csv`) already contains Shape CORE reconstructed curvature, bending angle, and 3-D shape, and not the underlying raw optical measurements.

Changes made to the article Data Records, Section “File naming and outputs” was revised.

Comment 5.40 *The statement that processed FBG quantities are retained “without further modification” is inaccurate because units, coordinate frames, timestamps, filtering, and sampling are modified during post-processing.*

Response to Comment 5.40 We agree; this statement was inaccurate. As the reviewer notes, and as Table 3 itself shows (curvature reported in mm^{-1} in the raw file versus m^{-1} in the processed file), the FBG curvature and bending angle are unit-converted, timestamp-corrected, low-pass filtered, and resampled identically to the other processed streams; the reconstructed 3-D shape additionally undergoes a coordinate-frame change into the robot body frame during this processing. We have corrected the text accordingly.

Changes made to the article Data Records, Section “File naming and outputs” was revised.

Comment 5.41 *Several additional code/documentation inconsistencies should be cleaned up, including incorrect torque units in plot labels, obsolete function calls, inconsistent comments on FBG column layout, and incomplete MATLAB/Python dependency documentation.*

Response to Comment 5.41 We thank the reviewer and have audited the release against each point. Units in plot labels, comments describing the code behavior and obsolete function calls were identified and removed. We also removed several test scripts that were not part of the acquisition pipeline used to record the dataset and had no documented dependencies. The software and hardware dependency list are provided in the `DEPENDENCIES.md` file.

Closing summary and recommendation

Reviewer summary *Overall, the paper addresses an existing research gap in the field of soft robotics. The primary motivation and the targeted scientific contribution are clear and valid. However, the paper and the dataset need more thoroughness and carefulness to reliably deliver the intended contribution and allowing others to use the dataset. A more precise use of terms and definitions is required in the manuscript. Also, several inconsistencies between the manuscript, the released acquisition/processing code, and the documented data structure need to be resolved. These include the interpretation of processed motion-capture data, calibration procedures, trajectory generation, temporal synchronization, filtering, and the reproducibility of the released processing pipeline. The dataset appears promising, but these issues should be corrected to ensure reliable third-party reuse.*

Reviewer recommendation *A major revision is necessary before the paper and the dataset can be published.*

Response to reviewer summary We thank the reviewer for this thorough and careful review. Addressing these comments strengthened the technical validity of both the manuscript and the released dataset, and identified several inconsistencies between the manuscript, the code, and the data that we would not otherwise have caught. We believe the revised manuscript and dataset address the concerns raised.

References

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